Multivariate Distributions and Processes: Recap

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Factoring a joint distribution

Any joint density factors into a product of conditionals:

$$f(x_1, \ldots, x_d) = f(x_1)\, f(x_2 \mid x_1) \cdots f(x_d \mid x_1, \ldots, x_{d-1})$$

- **Sampler**: draw X_1, then X_2 given X_1, and so on. Each step is a univariate problem from days 1–3.
- Any ordering of the coordinates works.
- **Checking**: a QQ plot checks one coordinate at a time, and would also pass if the coordinates were drawn independently. Check the dependence too, e.g. with a correlation.

Multivariate normal

If $\boldsymbol{Z} \sim N_d(\mathbf{0}, \boldsymbol{I}_d)$ and $\boldsymbol{\Sigma} = \boldsymbol{C}\boldsymbol{C}^T$, then

$$\boldsymbol{C}\boldsymbol{Z} + \boldsymbol{\mu} \sim N_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

- The **Cholesky factorization** gives a lower triangular $\boldsymbol{C}$. In R, `chol(Sigma)` returns $\boldsymbol{C}^T$, so use `C <- t(chol(Sigma))`.
- Cholesky is factoring. With unit variances and correlation ρ:

$$X_1 = Z_1, \qquad X_2 = \rho X_1 + \sqrt{1-\rho^2}\, Z_2,$$

so X_2 is drawn from its conditional $N(\rho X_1, 1-\rho^2)$.

Application: correlated tests

m test statistics on the same subjects, $\boldsymbol{Z} \sim N_m(\mathbf{0}, \boldsymbol{\Sigma})$ under the null.

- **Bonferroni** rejects when $|Z_i| > \Phi^{-1}(1 - \alpha/2m)$. Its FWER is at most α, and far below it when the tests are correlated.
- **Max-$|Z|$**: reject when $|Z_i|$ exceeds the 95th percentile of $\max_i |Z_i|$, estimated by simulating $\boldsymbol{Z}$. Its FWER is 5%.

Blue crabs, five size measurements, correlations above 0.95:

	Bonferroni	max-$\lvert Z \rvert$
cutoff	2.58	2.10
body depth, $z = 2.57$	not significant	significant

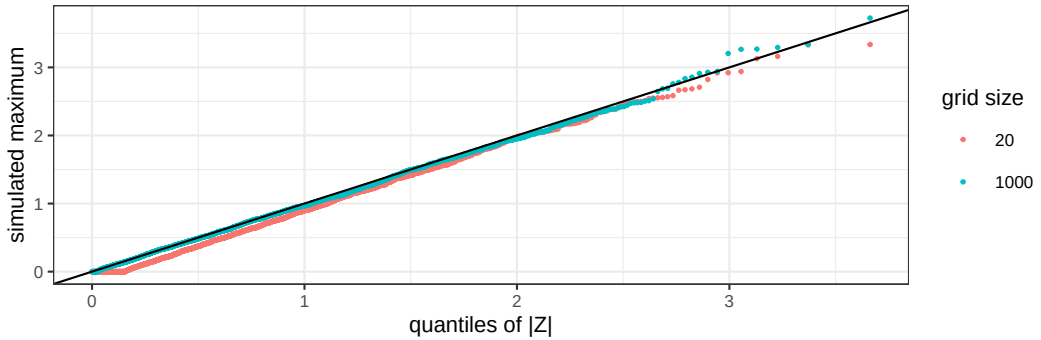
Stochastic processes

A process $\{X(t)\}$ is simulated at finitely many times, one time at a time, by factoring.

- **Random walk**: $X(t) = X(t-1) \pm 1$. It depends on the past only through $X(t-1)$, which makes it a **Markov chain**.
- **Brownian motion**: on a grid, $W(t_i) \mid W(t_{i-1}) \sim N(W(t_{i-1}), t_i - t_{i-1})$.
- **Brownian bridge**: $B(t) = W(t) - t\,W(1)$, Brownian motion conditioned on $W(1) = 0$.
- **Poisson process**: $\mathrm{Exp}(\lambda)$ gaps by inversion (day 1), arrival times as sums of gaps (day 2), and $N(t) \sim \mathrm{Poisson}(\lambda t)$.

Failure case: a coarse grid

The maximum of W on $[0, 1]$ has the distribution of $|Z|$ (the reflection principle).



Every grid value is an exact draw of $W(t_i)$, but the peaks fall between grid points, so the maximum on a coarse grid is too small.